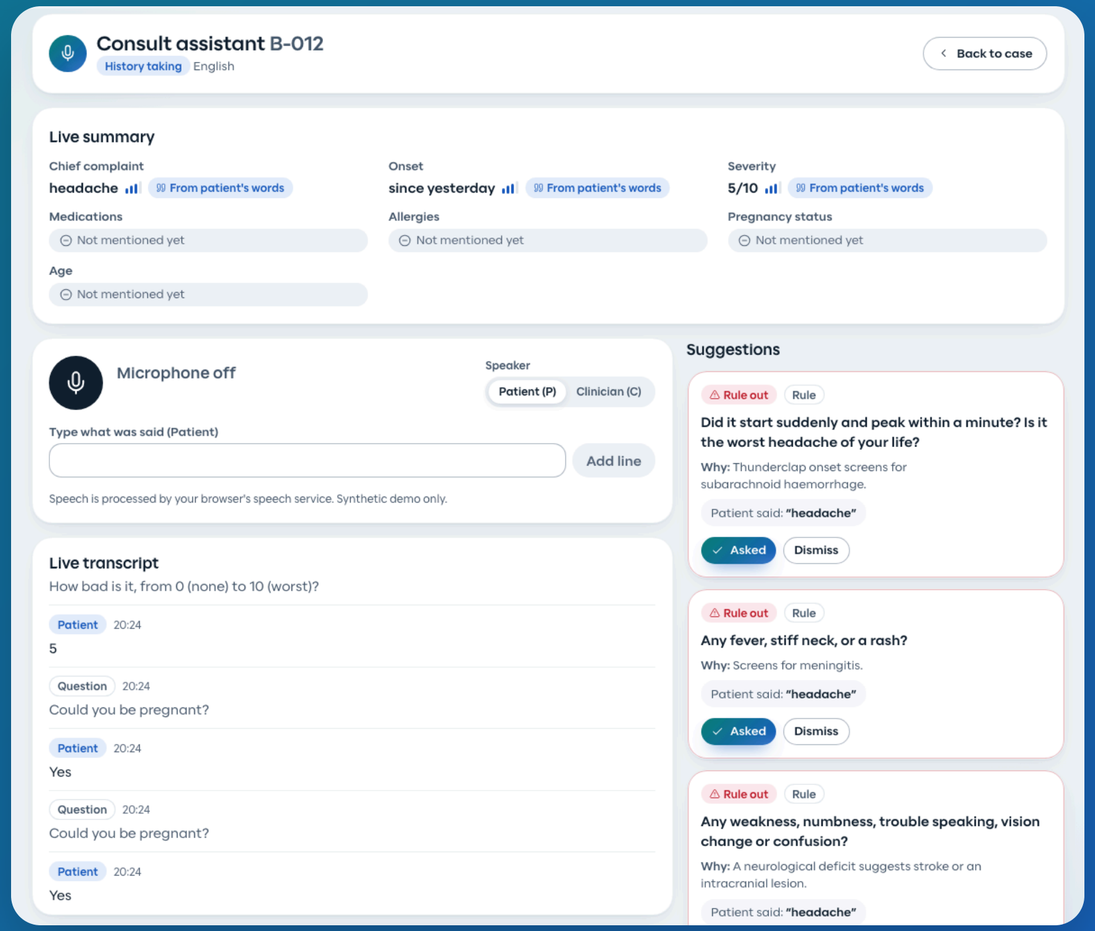
MedX

One patient journey, from the first words to the pharmacy. Every field points back to the patient's own words.



Meet Fin, the MedX dolphin. Fin listens closely and stays a helper; the people decide.



Research prototype. Supports clinicians; the AI never diagnoses, orders, prescribes or confirms triage.
Synthetic data only. github.com/KingPhuripol/MedX-AI

A language gap and a wrong medicine list are both safety events

49.1% vs 29.5%

Physical harm in adverse events, limited-English patients vs English speakers, six US hospitals. Communication errors caused 52.4% vs 35.9%.[1]

13–59%

Patients with at least one unintended error in their admission medication list across six studies; omission was the commonest type.[2]

49.2% vs 27.0%

Doctor's day on EHR and desk work vs face time with the patient, four specialties.[3]

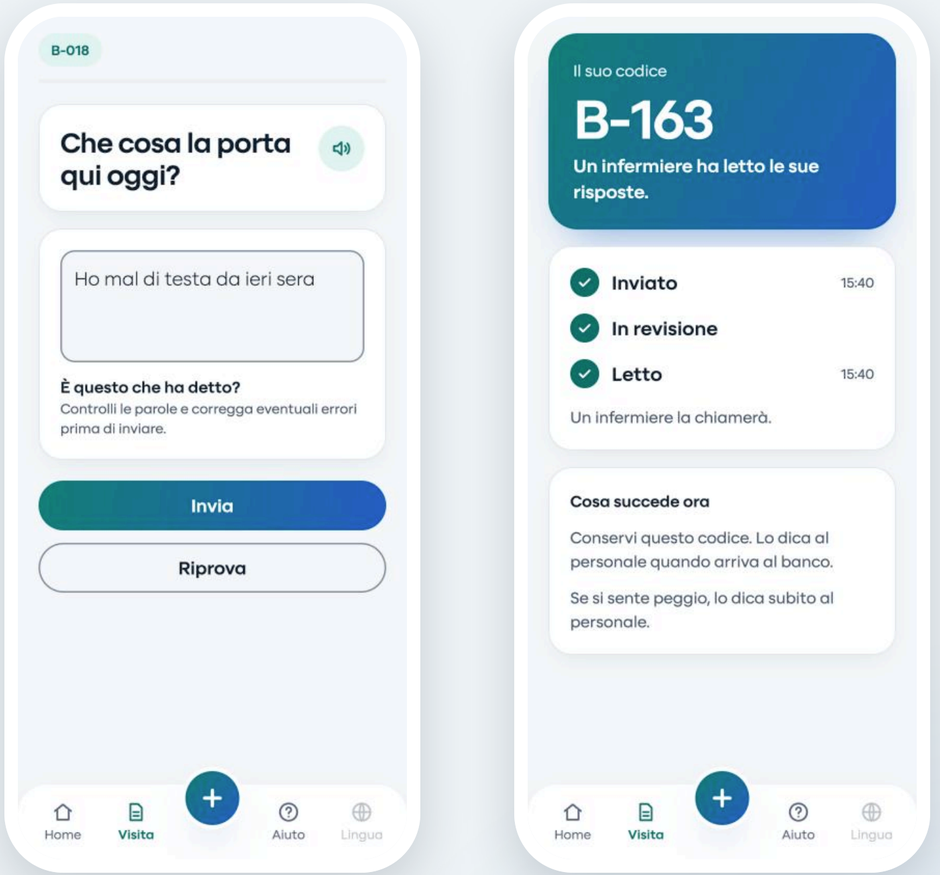
Thailand had 32.97 million foreign tourist arrivals in 2025.[4] At the desk, the patient and the staff often do not share a language.

1 Divi et al., Int J Qual Health Care 2007 (PMID 17277013). 2 Studies M1 to M6 in docs/research/PHARMACY-PROBLEM.md; settings differ, so a range is shown. 3 Sinsky et al., Ann Intern Med 2016 (PMID 27595430). 4 Ministry of Tourism and Sports via Bernama, 6 Jan 2026. Full list: docs/research/PROBLEM.md.

One patient journey, four people

1 Patient	Speaks or types in Thai, Italian or English. The patient checks the voice transcript before sending.
2 Nurse	History-taking with a consult assistant, vitals, and a triage suggestion the nurse confirms.
3 Doctor	Considerations for clinical judgement, a draft note to sign, and orders with automatic checks.
4 Pharmacist	Verifies or returns each prescription, and reconciles the medicines the patient named.

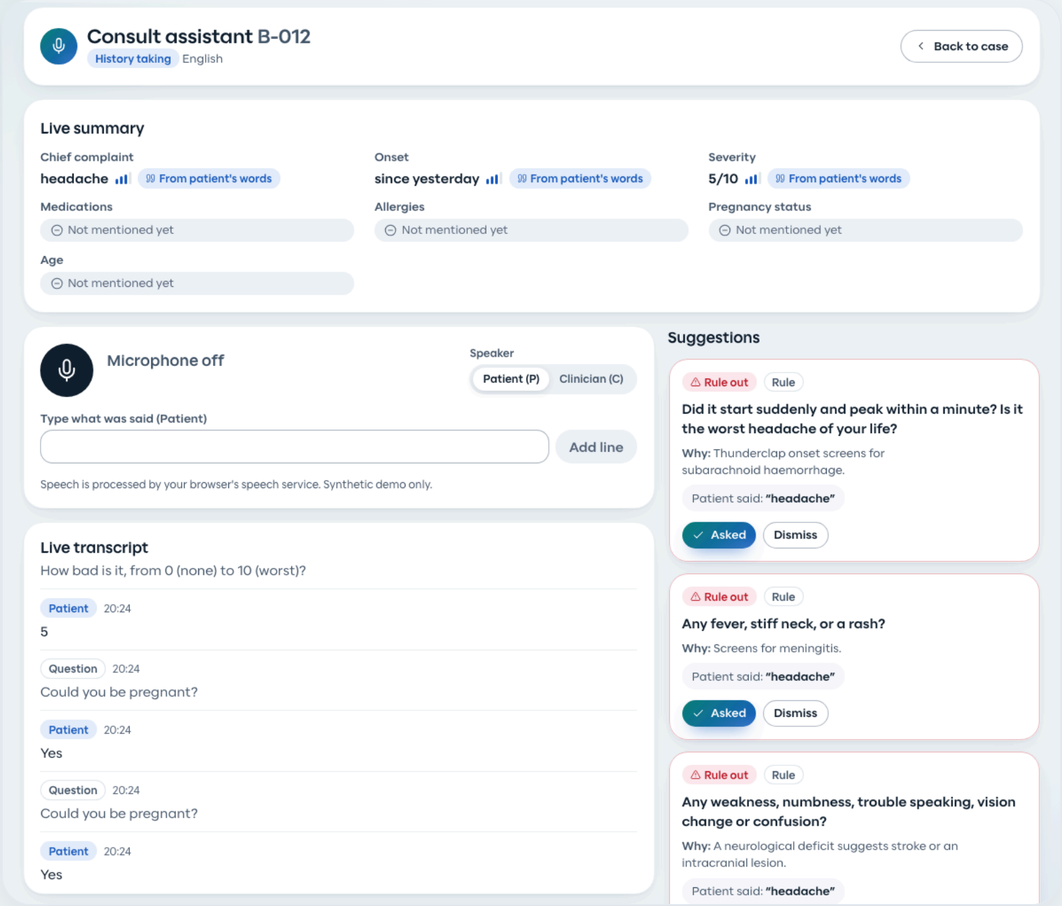
A clinic manager sees the queue, escalations and abstain rate. At every step a human decides; the AI never diagnoses, orders, prescribes or confirms triage.

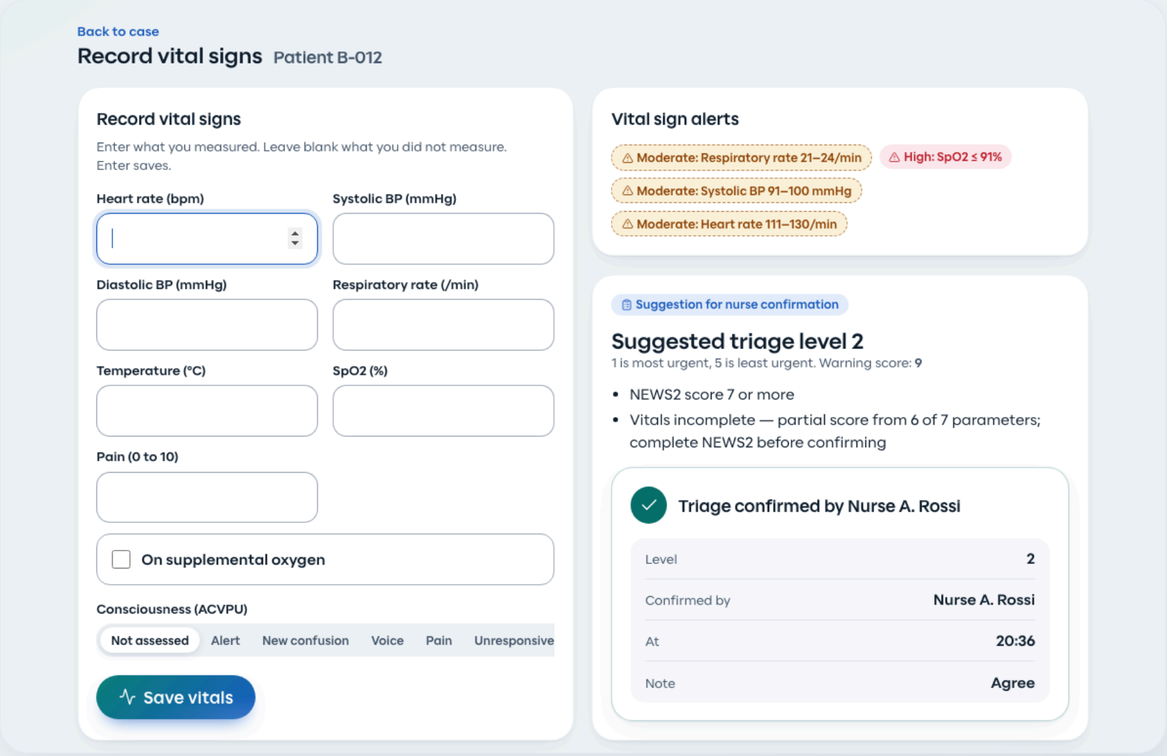


Left: the patient confirms what the voice agent heard. Right: the patient's phone shows when a nurse has read the answers.

Real screenshots from the built app (Expo patient app on web, Italian UI; English is the primary demo language). Voice input uses the phone's speech service.

Nurse: ask the next question, then confirm the triage

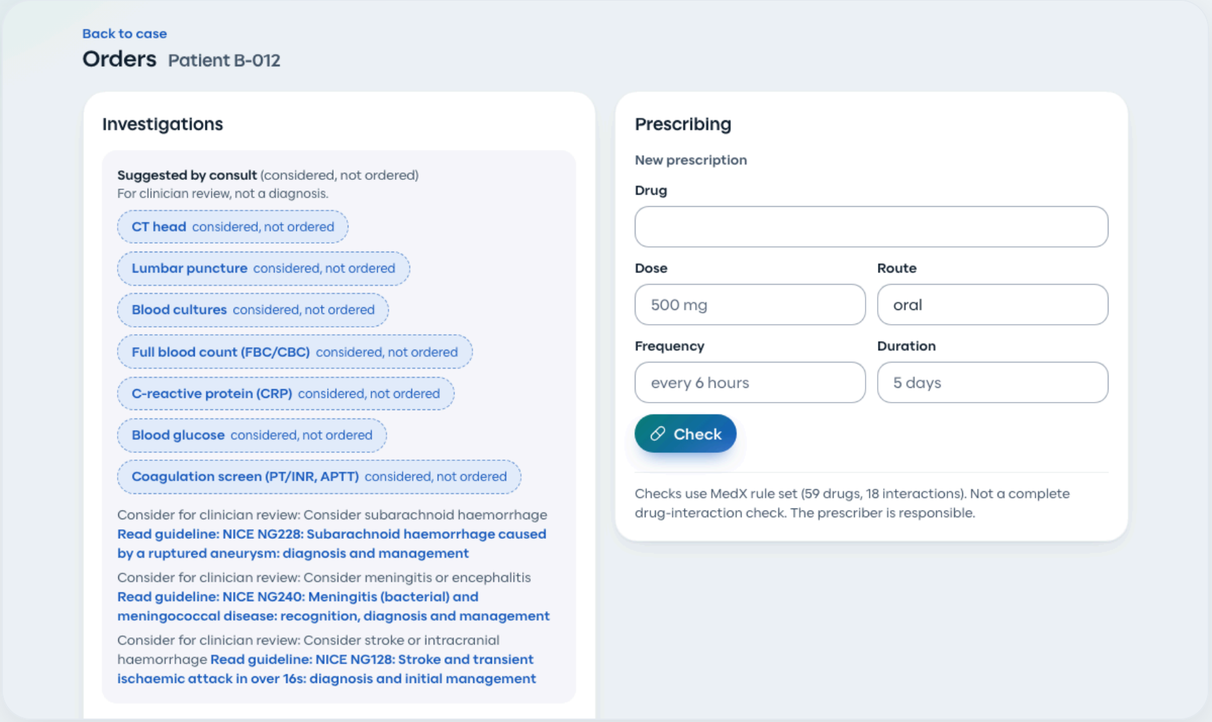




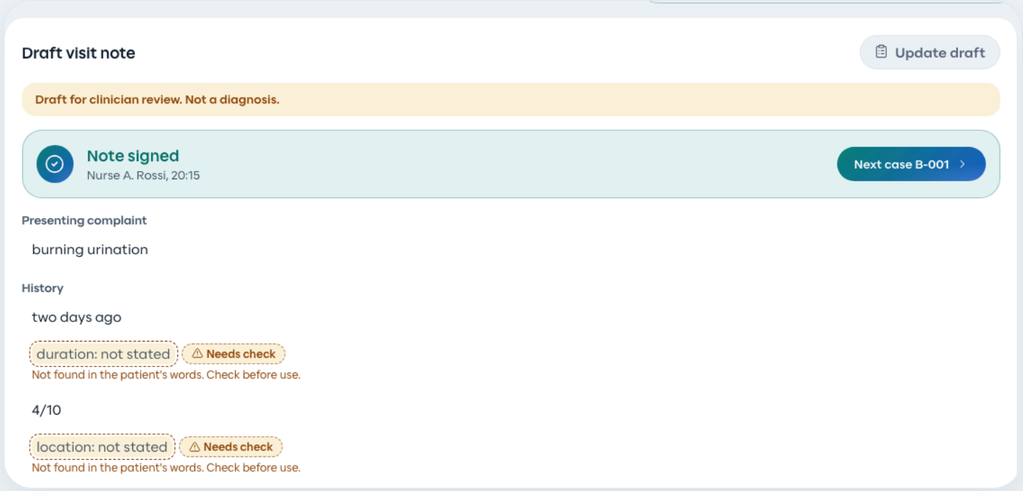
Left, consult assistant: it listens, keeps a live summary where each field links to the patient's words, and pops up "ask next" and "rule out" questions. **Right, vitals:** a NEWS2-based triage level is only a suggestion; the nurse confirms it.

Screens: nurse consult assistant and vitals, synthetic case B-012, built app. Speech is processed by the browser speech service. Triage level is a suggestion that the nurse confirms.

Doctor: considerations, a draft note, orders with checks



Considered, not ordered: workup ideas and guideline links appear for the doctor only, labelled "for clinician review, not a diagnosis". The AI never orders; the doctor does.

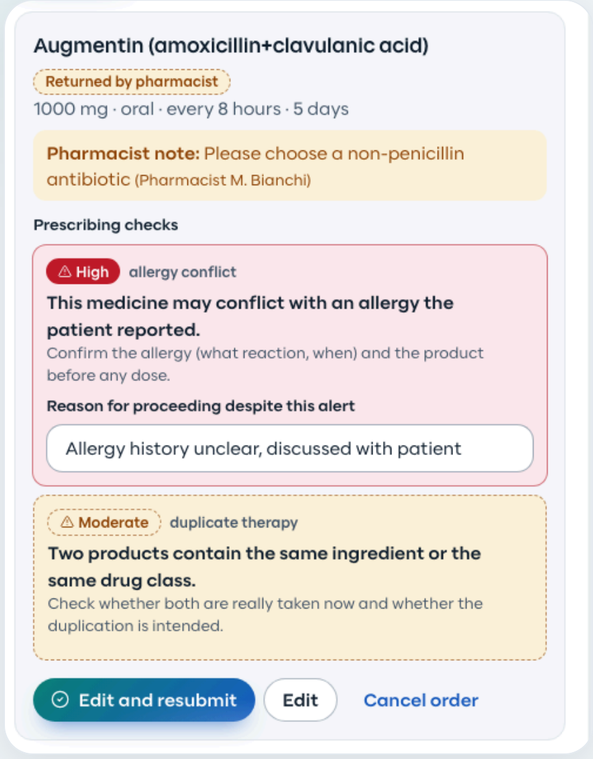


Draft visit note: built from the patient's words; anything not found shows "Needs check". The doctor reviews and signs.

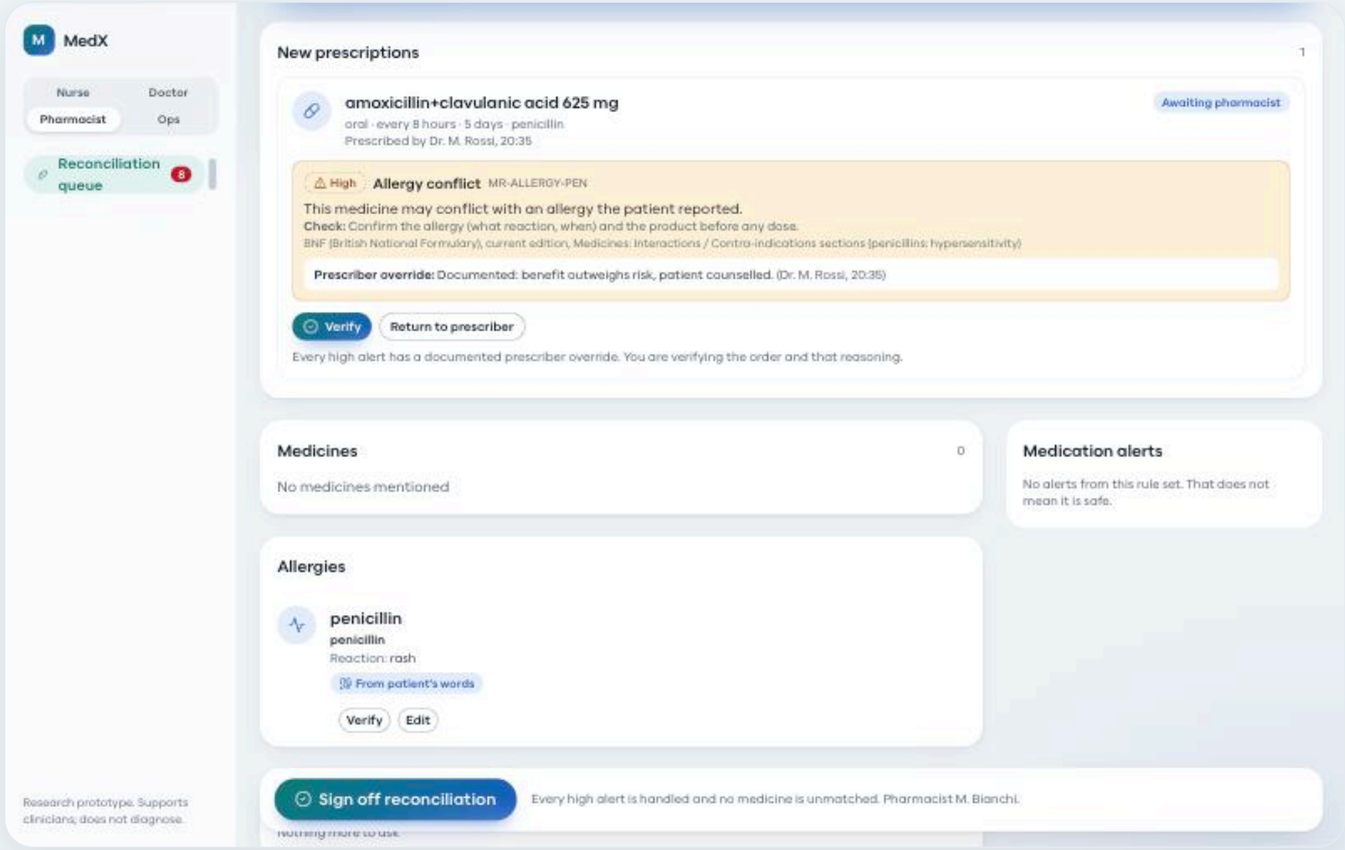
Next: orders. Prescriptions are checked by rules, not by the AI, and are shown on the next slide.

Screens: doctor orders view and draft note, synthetic cases, built app.

Prescribing checks, an override reason, a pharmacist loop



Doctor: rule-based checks (allergy, interaction, duplicate, pregnancy, dose range); a high alert needs a written reason.

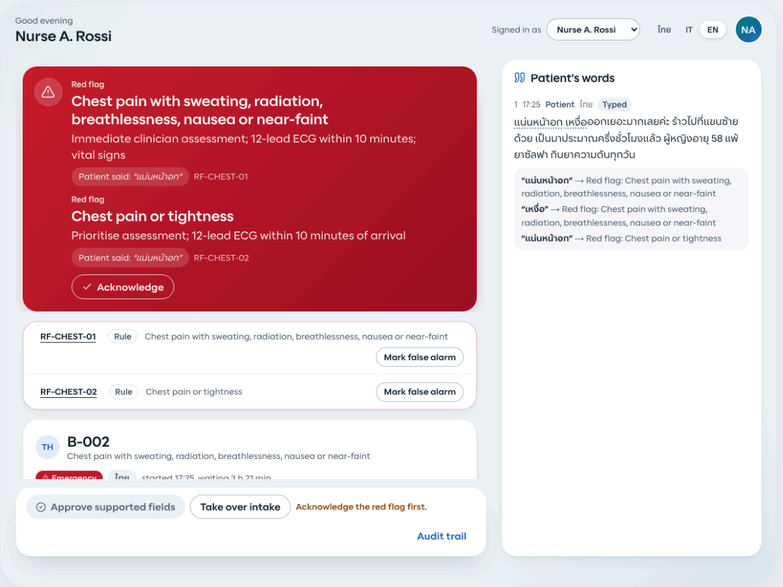


Pharmacist: sees the alert and the doctor's reason, then verifies or returns the order with a note. Returned orders come back to the doctor to edit and resubmit.

Screens: doctor prescribing view and pharmacist queue, synthetic cases, built app. Checks use a small rule set (59 drugs, 18 interactions), not a complete drug-interaction check.

Safety by design, in five answers

Wrong or unsupported answers	No verified quote means "needs check". Quotes must match the patient's words. Injection handling is not yet evaluated end to end.
Sensitive information	Pattern-based redaction before storage or any model call; imperfect. Synthetic data only. Keys stay server-side.
Uncertain results	Confidence on every field. Low confidence or provider failure: abstain, "needs human review".
High-risk decisions	Red-flag rules run before any model. The Gemma screen is add-only: "model-suggested, confirm". Triage is confirmed by the nurse; orders are entered by the doctor.
Human review	Nothing is final until a clinician approves. Overrides need a reason. Every action is in the audit log.



Thai patient, chest pain and sweating in the first message: the nurse screen is interrupted and the patient's sentence is highlighted. The patient's phone shows a calm "go to the desk now" screen.

Limits: doctor-only "consider" items come from a fixed 12-presentation list, not a differential. NEWS2 triage is an adult suggestion the nurse confirms. Prescribing checks use a small rule set (59 drugs, 18 interactions). Speech goes to a cloud speech service. Redaction is imperfect; reviewer identity is self-declared; Gemma-on-AMD is unmeasured (mock model). Paths: pitch/SAFETY.md.

Evidence we can show, and the gap that motivated the screen

What	Count	Where
Synthetic cases (40 intake + 12 medication)	**52**	data/cases
Blind holdout cases, written without seeing the rules	**18**	data/holdout
Red-flag rules / reference tests	**18 / 175**	docs/clinical
Drugs in dictionary / reference tests	**59 / 194**	docs/clinical
Automated tests, all passing	**1,382**	apps/web, vitest

Small synthetic blind holdout (18 cases), keyword rules alone

7 of 10

emergencies escalated; 3 of 8 non-emergencies raised a false alarm. Misses were lay wording keyword lists cannot cover. That is why Gemma adds a safety screen.

Designed to run Gemma on AMD via the hackathon's AMD Token Factory (OpenAI-compatible). Until the endpoint is connected, current results use a mock model.

With Gemma via AMD Token Factory

Holdout recall	Precision	Latency p50 / p95
[pending AMD run]	[pending AMD run]	[pending AMD run]

Synthetic evaluation, research prototype, small n; not clinically validated. Holdout: eval/results/holdout-rules-only.md, recorded before any change. Test count: npx vitest run in apps/web, 17 Oct 2026. Pre-declared metrics: docs/ai/METRICS.md.

Business case: who buys, and why

Buyer: private hospitals and clinics with international patients, and clinics in tourist areas (Thailand had 32.97M foreign arrivals in 2025[1]).

CEO / CFO	Interpreter cost and patient throughput at the front desk.
Head nurse	Documentation time: nurses spent 20.9% to 28.1% of a shift documenting in one observed study. [2]
Doctors	AI they can verify: each field links to the patient's sentence; nothing is final until the doctor signs.
Quality and risk team	Medication errors: 13% to 59% of patients had an admission-list error across six studies. [3]

Throughput

One structured intake in the clinician's language.

Safety

Red flags, prescribing checks, pharmacist loop, audit log.

Trust

Evidence-linked fields, honest abstain and "needs check".

Next step: pilot

One clinic, measured before and after. ROI figures to come from the pilot, not from this deck.

1 Ministry of Tourism and Sports via Bernama, 6 Jan 2026. 2 J Clin Nurs 2021 (PMID 33590554), ward nurses in one hospital. 3 Studies M1 to M6, docs/research/PHARMACY-PROBLEM.md; settings differ. No savings figure is claimed; none has been measured.

Next: prove it with clinicians before anyone relies on it

- Run the pre-declared evaluation on Gemma via AMD Token Factory and publish every result, including misses.
- Native-speaker review of Thai and Italian text.
- A timed pilot with nurses, doctors and pharmacists, then ethics approval before any real data.
- Not a medical device. Supports clinicians; the AI never diagnoses, orders, prescribes or confirms triage.

Open source, MIT.
github.com/KingPhuripol/MedX-AI

Team MedX. Built by a human lead orchestrating Claude (Opus/Sonnet) and OpenAI Codex agents in role-based lanes.



Fin listens closely.
The people decide.